1. Basic Concept in Geometry.

1. Where did Geometry come from?

The word Geometry comes from two Greek words: 'Geo' (which means Earth) and 'Metria' (which means measuring). So, it started because people needed to measure land!

History Tip:

- Ancient Egyptians used Geometry to build the massive pyramids 3,000 years before the Christian Era.
- In India, during the Vedic period, people used geometrical shapes to build altars for worship.
- Great Indian mathematicians like Aryabhat, Varahamihir, and Bramhagupta contributed a lot to this subject.

2. The Building Blocks: Point, Line, and Plane

In Geometry, there are three things we don't strictly define, just like we don't define numbers:

- **Point:** A tiny dot.
- **Line:** In this book, "line" always means a straight line. It is actually a collection of many points.
- **Plane:** A flat surface (like the top of a table) that goes on forever.

Daily Life Example:

Think of a tiny grain of sand as a point, a long straight piece of thread as a line, and the surface of a calm pond as a plane.

3. Co-ordinates and Distance

Every point on a number line is linked to a number. This number is called its **Co-ordinate**.

How to find the distance between two points:

To find how far apart two points are, you simply subtract the smaller number from the larger number.

The Formula:

$\text{Distance} = \text{Larger Co-ordinate} - \text{Smaller Co-ordinate}$

What the symbols mean:

- $d(A, B)$ means the distance between point A and point B.
- $l(AB)$ means the length of the segment connecting A and B.

Why use it?

We use this to find out exactly how long a path is between two spots on a map or a ruler.

Important Exam Points:

- Distance is always a positive number (or zero if the points are the same). You can never have a distance of -5 cm!

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- $d(A, B)$ is the same as $d(B, A)$.

Daily Life Example:

If you are standing at the 2-meter mark on a track and your friend is at the 10-meter mark, the distance between you is $10 - 2 = 8$ meters.

4. Betweenness (Who is in the middle?)

If three points (let's call them P, Q, and R) are on the same straight line, they are called collinear points. One of them must be between the other two.

- **The Rule:** Point Q is between P and R only if the distance from P to Q plus the distance from Q to R equals the total distance from P to R.
- **Symbol:** We write this as P - Q - R.
- **Formula:** $d(P, Q) + d(Q, R) = d(P, R)$.

Daily Life Example:

If you travel from your house (A) to school (C), and you stop at a bakery (B) on the way, the bakery is "between" your house and school if the distance House-Bakery + Bakery-School = total House-School distance.

5. Segments, Rays, and Lines

It is easy to get these mixed up, so let's look at them clearly:

- **Line Segment (seg AB):** It has a start (A) and an end (B). It has a fixed length.
- **Ray (ray AB):** It starts at one point (A) and goes on forever in one direction (towards B and beyond). Point A is called the starting point.
- **Line (line AB):** It goes on forever in both directions.

6. Important Properties of Segments

- **Congruent Segments:** If two segments have the same length, they are called congruent (symbol: $\cong$).
- **Midpoint:** A point that sits exactly in the middle of a segment, dividing it into two equal parts. Every segment has only one midpoint.
- **Perpendicular Segments:** When two segments or rays meet at a 90° angle (like the corner of a square).

7. Conditional Statements and Proofs

This is where we learn to think like a detective!

Conditional Statements:

These are "If... then..." statements.

- **Antecedent:** The part after "If" (The things we already know).

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- **Consequent:** The part after "then" (The thing we want to prove).
- **Converse:** When you swap the "If" and "then" parts. **Note:** The converse is not always true!

How do we prove things?

- **Postulates:** Simple truths that we accept without needing proof (e.g., "There are infinite lines passing through a point").
- **Theorems:** Properties that are proved logically using steps.
- **Proof:** The logical argument used to prove a theorem. It can be Direct (straightforward steps) or Indirect (assuming the opposite and showing it leads to a mistake).

Summary for Quick Revision

- Geometry means Earth-measuring.
- Distance is always (Larger Co-ordinate - Smaller Co-ordinate) and is never negative.
- Betweenness (A-B-C) means $d(A, B) + d(B, C) = d(A, C)$.
- A Segment has two ends; a Ray has one; a Line has none.
- Congruent means equal in length.
- Conditional statements use "If-Then".

Key Terms & Meanings

- **Collinear Points:** Points that lie on the same straight line.
- **Co-ordinate:** The number associated with a point on a number line.
- **Congruent:** Exactly the same size/length.
- **Midpoint:** The exact center point of a segment.
- **Theorem:** A mathematical statement that has been proven true.

Exam-Oriented Questions & Answers

Q1. The co-ordinates of point A and B are -2 and 5. Find $d(A, B)$.

Answer: Larger number is 5, smaller is -2.

$d(A, B) = 5 - (-2) = 5 + 2 = 7$.

The distance is 7 units.

Q2. If $d(P, Q) = 3$, $d(Q, R) = 5$ and $d(P, R) = 8$, which point is between the other two?

Answer: Since $3 + 5 = 8$, we see that $d(P, Q) + d(Q, R) = d(P, R)$.

Therefore, Point Q is between P and R (P-Q-R).

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Q3. Define a Midpoint.

Answer: If A-M-B and segment AM $\cong$ segment MB, then M is called the midpoint of segment AB.

Q4. What is a Converse statement? Give an example.

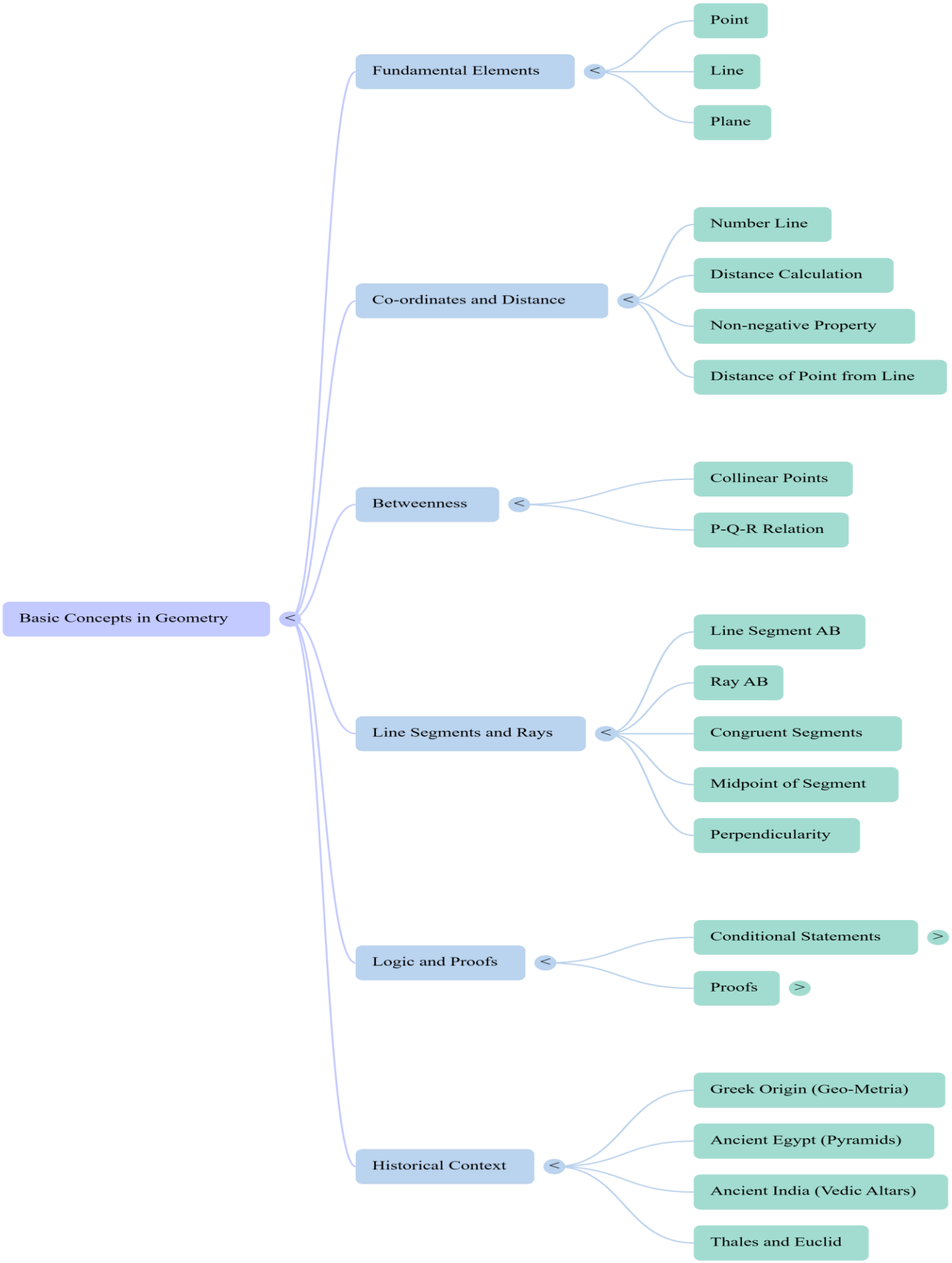
Answer: A converse is formed by interchanging the antecedent and consequent of a conditional statement.

Example: Statement: "If a quadrilateral is a rhombus, then its diagonals are perpendicular." Converse: "If the diagonals of a quadrilateral are perpendicular, then it is a rhombus".

Q5. State one of Euclid's postulates.

Answer: There are infinite lines passing through a point (OR: There is one and only one line passing through two points).

Q6. Is distance ever negative? Why?

Answer: No. The distance between any two distinct points is always a positive number because it is calculated by subtracting the smaller co-ordinate from the larger one.

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